Detecting departure and destination aerodromes from ADS-B

Version 6 — the consolidated study, and what to run in production

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Summary

OPDI derives a flight's departure and destination aerodrome from ADS-B state vectors alone. This page consolidates six versions of that work into one account: what the three candidate methods are, how each behaves across its whole parameter space, which settings are best for departures and for arrivals, what the ideas that did *not* work were, and what the pipeline should ship.

The short version is in three parts.

The two roles want different algorithms. Departures are best served by `endpoint` — take the aerodrome nearest the first fix, but only commit when the fix is close and low, and abstain otherwise. Arrivals are best served by `trend`, the algorithm already in production. That split holds at every exchange rate tested.

Both roles improve. The recommended configuration — `endpoint` for departures, exact-distance `trend` for arrivals — scores **133,781** against production's **123,664** on a three-day sample of 95,116 flights: a gain of **+10,117**, every figure from a real pipeline run.

Arrivals improve too, but only after a defect in the pipeline was fixed — and that is the result worth carrying away. Sweeping `trend` said its never-tuned flight-level gate should move from FL40 to FL60, worth over a thousand flights. Run through the pipeline, the same change *lost* ground.

The cause was not the tuning. The sweep ranked candidate aerodromes by exact distance; the pipeline kept only the candidates at the minimum **H3 ring count** — an integer stepping about 5.2 km at resolution 7 — and measured only among those. An aerodrome a kilometre further out was discarded before its distance was ever computed. Raising the flight-level gate admits

samples from higher up, where more aerodromes fall in the same ring, so the coarser rule had more work to do exactly where it was least able to do it.

The sweep was not over-estimating. It was modelling a better algorithm than production implemented. With ranking on exact distance, every tuned arrival value gains ground, and the pipeline's own optimum turns out to be the one the sweep had been pointing at all along.

> **i** Reading this page
>
> Every number below is an inline expression over a CSV in `data/`, and every one of those CSVs is produced by a job in `benchmarks/regenerate_v6.py` — which this page runs before drawing anything. Nothing is typed into the prose, and nothing is read that the checked-out code did not produce. Section 16 lists the command and commit behind each figure. Where this page disagrees with versions 1–5, it says so and gives the reason; those versions stay published unchanged.

1 How the metrics work

Five quantities are reported for every setting, for each role separately. Four of them are unavoidable and the fifth is a judgement, so it is worth being precise about all five before any of them are used to argue for a parameter.

Write N for the number of ground-truth flights in the sample, a for the number the method *answers* (names an aerodrome, or explicitly says out-of-area), c for the number it answers **correctly**, and $w = a - c$ for the number it answers **wrongly**.

Coverage is a/N: the share of flights the method is willing to commit on. A method that abstains everywhere has coverage zero and cannot be wrong.

Accuracy is c/a: of the flights it committed on, the share it got right. Accuracy says nothing about how often the method declined to answer, so a method can buy accuracy simply by abstaining more.

Correct (c) and **wrong** (w) are the raw counts. They matter because coverage and accuracy generally move in opposite directions — every extra answer a method gives is one it was previously unsure about, so raising coverage tends to lower accuracy — and neither ratio says whether a particular exchange was worth making. The counts do.

Score is $c - k\,w$. It exists to make the exchange rate explicit: a change to a parameter is worth taking if and only if it wins more than k correct answers for each wrong one it introduces.

The value of k is where judgement enters. This study uses $k = 2$ by default, on the following argument. A **silence** — an abstention — corrupts one movement count, and it is *visible*: it arrives downstream as an explicit null that any consumer can filter on. A **wrong aerodrome** corrupts *two* movement counts, the one it invents and the one it omits, and it is *invisible*: nothing downstream can distinguish it from a correct answer. Two errors for one, plus undetectability, is the case for $k = 2$.

That is an argument, not a measurement, and rankings do move with k. So every headline choice on this page is also scored at $k = 1$ and $k = 3$, and where the ranking changes, the page says so.

1.1 Out-of-area, and why the headline numbers are decomposed

Roughly one flight in six in the sample genuinely begins or ends outside the observed area. Such a flight has no nameable aerodrome, and the correct answer is the literal label `OOA`. This creates a way to score well for a bad reason: a method that labels liberally out-of-area collects those flights for free.

So each side is also reported three further ways — out-of-area recall and precision, which show whether the free wins are being taken honestly, and in-area coverage and accuracy, which restrict to flights whose truth is a real aerodrome and are the numbers to read when comparing naming rules.

Table 1: The reported quantities.

Quantity	Definition	Reads as
Coverage	a/N	willingness to commit
Accuracy	c/a	reliability when committed
Correct	c	flights named right
Wrong	$w = a - c$	flights named wrong
Score	$c - k\,w$	net value at exchange rate k
OOA recall	OOA hits / OOA in truth	does it recognise what it cannot name?
OOA precision	OOA hits / OOA predicted	is it over-using the OOA label?
In-area coverage	answered / truth is a real aerodrome	commitment with the free wins removed
In-area accuracy	correct / answered, truth a real aerodrome	reliability with the free wins removed

2 The sample

The 2025 period is 5-7 June 2025, three consecutive days, giving **95,116 ground-truth flights** after alignment with the Network Manager reference.

Ground truth is the EUROCONTROL Network Manager flight table, joined to ADS-B tracks on aircraft address, callsign and date. The join key is `AIRCRAFT_ADDRESS`, which is the `icao24` of the ADS-B message.

3 The three methods

3.1 `trend` — what production runs today

For every state vector below a flight-level cap and within a radius of an aerodrome, `trend` asks whether smoothed barometric altitude is rising or falling. Each vector casts a vote. If climbing votes exceed descending votes by more than a margin, that aerodrome is the departure; if the reverse, the destination. The nearest surviving candidate wins.

Two of its four constants had never been swept before this study: the flight-level cap, fixed at FL40, and the detection radius, fixed at 30 NM.

3.2 `nearest` — the *traffic* library's rule

The reference implementation in the `traffic` library takes the aerodrome nearest the first and last point of the trajectory, unconditionally. It never abstains, so its coverage is total and its accuracy is whatever the geometry gives. It is included as the external baseline.

3.3 `endpoint` — `nearest`, with abstention

`endpoint` is `nearest` plus two gates: the endpoint must lie within a radius of the aerodrome, and either on the ground or below a height above the field. Failing both, the method abstains rather than guessing. A scheduled-service penalty breaks ties toward aerodromes with commercial traffic.

4 Trend: the full parameter sweep

4.1 The untuned baseline

- **ADEP** at the production setting (FL40, margin 4, 30 NM, no penalty): coverage 68.35%, accuracy 98.72%, 64,179 correct, 833 wrong, score 62,513.
- **ADES** at the production setting (FL40, margin 4, 30 NM, no penalty): coverage 67.39%, accuracy 97.51%, 62,507 correct, 1,593 wrong, score 59,321.

4.2 The best setting found

- **ADEP**: FL120, margin 0, 20 NM, penalty 10 NM — coverage 77.43%, accuracy 98.59%, 72,605 correct, 1,039 wrong, score 70,527 (+8,014 against the untuned baseline).
- **ADES**: FL60, margin 2, 20 NM, penalty 10 NM — coverage 69.51%, accuracy 97.46%, 64,431 correct, 1,681 wrong, score 61,069 (+1,748 against the untuned baseline).

4.3 The full trend grid

`trend` has three geometric parameters, so the sweep is a cube and cannot be printed as one table. It is sliced instead: for each role, score by flight-level cap against radius at that role's best vote margin, then the margin's own profile, then the penalty. Every cell of the cube is in `data/trend_sweep_2025.csv`.

ADEP — score at $k = 2$, by flight-level cap (rows) and radius in NM (columns), at the winning vote margin of 0

FL cap	5	10	15	20	30	40	60	80
20	56,083	55,987	55,921	55,872	55,743	55,675	55,415	54,633
30	62,659	62,896	62,827	62,759	62,590	62,524	62,216	61,482
40	64,617	65,901	65,823	65,746	65,533	65,437	65,109	64,220
60	65,139	67,929	67,951	67,869	67,601	67,481	67,107	66,253
80	64,947	68,763	69,367	69,321	69,042	68,867	68,444	67,573
100	64,645	68,789	70,070	70,241	69,946	69,718	69,229	68,316
120	64,191	68,377	70,030	70,527	70,330	70,073	69,546	68,620
150	62,873	67,194	68,908	69,459	69,553	69,298	68,707	67,765

FL cap	5	10	15	20	30	40	60	80
200	59,828	63,880	65,262	65,608	65,696	65,494	64,826	63,851

ADES — score at $k = 2$, by flight-level cap (rows) and radius in NM (columns), at the winning vote margin of 2

FL cap	5	10	15	20	30	40	60	80
20	58,227	58,884	58,650	58,513	58,362	58,255	58,104	58,032
30	58,551	60,187	60,098	59,970	59,807	59,694	59,556	59,472
40	58,447	60,299	60,686	60,736	60,591	60,464	60,316	60,221
60	57,675	59,045	59,771	61,069	61,057	60,906	60,695	60,552
80	56,706	58,052	58,733	60,269	60,492	60,305	60,050	59,853
100	54,692	56,010	56,645	58,122	58,613	58,496	58,195	57,937
120	52,974	54,202	54,877	56,083	56,579	56,469	56,135	55,851
150	51,430	52,282	52,677	53,692	53,945	53,866	53,512	53,208
200	49,002	49,236	49,149	49,695	49,385	49,093	48,736	48,402

The vote margin, at each role's winning cap and radius:

Role	Margin	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.43%	98.59%	72,605	1,039	70,527
ADEP	2	77.19%	98.64%	72,422	995	70,432
ADEP	4	76.98%	98.69%	72,255	962	70,331
ADEP	8	76.32%	98.73%	71,671	925	69,821
ADEP	16	74.86%	98.79%	70,341	863	68,615
ADES	0	69.66%	97.37%	64,512	1,743	61,026
ADES	2	69.51%	97.46%	64,431	1,681	61,069
ADES	4	69.11%	97.53%	64,113	1,624	60,865
ADES	8	68.60%	97.61%	63,684	1,562	60,560
ADES	16	67.36%	97.70%	62,597	1,473	59,651

The scheduled-service penalty applied to `trend`, which production does not do at all, swept at each role's winning geometry:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.43%	96.39%	70,986	2,658	65,670
ADEP	5	77.43%	98.39%	72,457	1,187	70,083
ADEP	10	77.43%	98.59%	72,605	1,039	70,527
ADEP	15	77.43%	98.44%	72,497	1,147	70,203
ADEP	20	77.43%	98.29%	72,383	1,261	69,861
ADEP	30	77.43%	98.29%	72,383	1,261	69,861
ADES	0	69.51%	95.34%	63,033	3,079	56,875
ADES	5	69.51%	97.15%	64,230	1,882	60,466
ADES	10	69.51%	97.46%	64,431	1,681	61,069
ADES	15	69.51%	97.39%	64,386	1,726	60,934
ADES	20	69.51%	97.29%	64,322	1,790	60,742
ADES	30	69.51%	97.29%	64,322	1,790	60,742

Coverage is constant down each block and only accuracy moves, which is the signature of a pure re-ranking: the penalty changes which aerodrome is named, never whether one is.

5 Endpoint: the full parameter sweep

`endpoint` has two geometric parameters. The **radius** is how close to an aerodrome the trajectory's first or last point must be; the **height** is how far above field elevation it may be. Failing either, the method abstains.

The grid below is the whole sweep, scored for each role. The height row labelled *unrestricted* removes the height gate entirely, which collapses `endpoint` into `nearest` within the radius — so the last row of each table is the cost of having no abstention at all.

ADEP — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	28,787	28,715	28,689	28,665	28,660	28,652	28,652	28,652	28,652
500	39,417	39,321	39,263	39,225	39,220	39,210	39,210	39,210	39,210
1,000	52,473	52,363	52,281	52,235	52,231	52,211	52,211	52,211	52,211
2,000	62,143	62,014	61,914	61,849	61,839	61,793	61,789	61,789	61,789
5,000	66,127	68,510	68,363	68,256	68,179	68,067	68,040	68,040	68,040
8,000	66,263	70,073	70,656	70,465	70,355	70,239	70,204	70,202	70,202
10,000	66,247	70,285	71,758	71,571	71,417	71,271	71,222	71,220	71,218
15,000	66,221	70,296	72,586	72,737	72,522	72,241	72,182	72,176	72,174
20,000	66,187	70,160	72,244	72,355	72,140	71,751	71,659	71,641	71,639
25,000	66,141	69,991	71,766	71,455	70,608	69,803	69,639	69,612	69,606
30,000	66,110	69,865	71,150	70,168	68,777	67,467	67,255	67,148	67,141
35,000	66,039	69,232	69,345	66,928	64,180	61,581	61,213	60,975	60,932
40,000	66,003	68,922	68,200	64,986	61,106	57,099	56,343	55,812	55,589
unrestricted	66,004	68,909	68,161	64,919	61,004	56,962	56,190	55,639	55,406

ADES — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	32,794	32,700	32,581	32,435	32,360	32,270	32,254	32,252	32,252
500	48,209	48,106	47,932	47,769	47,692	47,597	47,581	47,575	47,575
1,000	52,503	52,375	52,153	51,958	51,866	51,755	51,731	51,725	51,725
2,000	53,539	54,536	54,300	54,069	53,942	53,816	53,774	53,766	53,766
5,000	53,498	55,613	57,173	56,941	56,723	56,511	56,408	56,400	56,400
8,000	53,583	55,639	57,685	57,785	57,483	57,240	57,109	57,097	57,097
10,000	53,590	55,585	57,645	57,954	57,671	57,401	57,262	57,250	57,250
15,000	53,618	55,498	57,124	57,112	56,756	56,418	56,265	56,251	56,249
20,000	53,675	55,392	56,373	55,329	54,666	54,219	54,040	54,002	54,000
25,000	53,798	55,432	55,937	54,358	53,407	52,565	52,286	52,230	52,224
30,000	53,878	55,382	55,450	53,357	51,915	50,639	50,101	50,010	49,998
35,000	54,005	54,813	53,206	49,372	46,498	43,197	41,678	41,034	40,828
40,000	54,792	55,169	52,361	47,109	43,022	37,204	34,968	33,307	32,735
unrestricted	54,946	55,306	52,460	47,132	42,888	36,857	34,583	32,870	32,282

5.1 Where each role's optimum sits

- **ADEP**: 30 NM, 15,000 ft — coverage 80.96%, accuracy 98.15%, 75,581 correct, 1,422 wrong, score 72,737.
- **ADES**: 30 NM, 10,000 ft — coverage 67.21%, accuracy 96.88%, 61,938 correct, 1,992 wrong, score 57,954.

Both optima are interior to the grid in both parameters, so neither is an artefact of where the sweep stopped.

6 The scheduled-service penalty

Both methods choose among surviving candidates by distance. Near a major aerodrome that often means choosing between the airport itself and a small airfield a few miles away that almost never sees commercial traffic. The penalty adds a fixed distance to every candidate without scheduled service, biasing the tie-break toward the aerodrome more likely to be the real one.

It is a pure re-ranking: it changes *which* aerodrome is named, never *whether* one is named. So coverage is flat across the whole sweep and only accuracy moves — which is exactly what makes it cheap.

Endpoint, at 40 NM and 15,000 ft:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	81.20%	97.30%	75,146	2,089	70,968
ADEP	5	81.20%	97.71%	75,461	1,771	71,919
ADEP	10	81.20%	97.97%	75,662	1,570	72,522
ADEP	15	81.20%	97.89%	75,604	1,627	72,350
ADEP	20	81.21%	97.67%	75,436	1,803	71,830
ADEP	30	81.20%	97.57%	75,357	1,876	71,605
ADES	0	69.99%	93.86%	62,480	4,090	54,300
ADES	5	69.98%	94.32%	62,780	3,784	55,212
ADES	10	69.96%	95.10%	63,280	3,262	56,756
ADES	15	69.91%	94.88%	63,098	3,402	56,294
ADES	20	69.90%	94.80%	63,030	3,454	56,122
ADES	30	69.81%	94.84%	62,978	3,423	56,132

7 How much the exchange rate matters

k is a judgement, so the honest thing is to show what it decides. The table below re-runs the argmax at $k = 1$, 2 and 3 for both methods and both roles.

k	Role	Best trend	trend score	Best endpoint	endpoint score	Winner
1	ADEP	FL120, margin 0, 20 NM, pen 10	71,566	40 NM, 20,000 ft, pen 10	74,229	endpoint
1	ADES	FL60, margin 2, 30 NM, pen 10	62,838	40 NM, 15,000 ft, pen 10	60,018	trend

k	Role	Best trend	trend score	Best endpoint	endpoint score	Winner
2	ADEP	FL120, margin 0, 20 NM, pen 10	70,527	30 NM, 15,000 ft, pen 10	72,737	endpoint
2	ADES	FL60, margin 2, 20 NM, pen 10	61,069	30 NM, 10,000 ft, pen 10	57,954	trend
3	ADEP	FL120, margin 0, 20 NM, pen 10	69,488	20 NM, 15,000 ft, pen 10	71,392	endpoint
3	ADES	FL40, margin 2, 15 NM, pen 10	59,494	20 NM, 8,000 ft, pen 10	56,349	trend

The **choice of method is the same at every k tried**: departures want `endpoint` and arrivals want `trend` whether a wrong answer is priced at one silence, two or three. That is the most robust finding on this page, and it does not depend on the judgement that fixes k.

The parameter *values* are less stable than the method choice, and in a direction that makes sense: a higher k prices wrong answers more dearly, so the argmax moves toward tighter gates that answer less often and are right more often when they do. Where a headline value moves with k, the justification table records it.

8 Negative results, re-measured

Ideas that were tried and did not work. All are re-run here against current code rather than quoted from the version that first reported them.

8.1 The approach cone

The box rule treats radius and height as independent, and they are not: an aircraft descending toward an aerodrome trades height for distance at roughly a constant rate. A 3° glideslope is 318 ft/NM and the "three times the flight level" rule of thumb about 333 ft/NM, so the endpoints that genuinely belong to an aerodrome should lie in a *cone*, not a box.

The two shapes disagree in both corners. Close and high is inside the box and outside the cone — an aircraft at 15,000 ft overhead is overflying, not arriving. Far and low is outside the box and inside the cone — an arrival whose reception dies at 8,000 ft and 30 NM is on profile.

Replacing the box with a cone is worse for both roles, at every slope swept.

- **ADEP**: best cone (1000 ft/NM, 40 NM) scores 65,738 against the box's 72,737, −6,999. It is worse on both counts — fewer correct *and* more wrong, so no exchange rate rescues it.
- **ADES**: best cone (500 ft/NM, 40 NM) scores 57,605 against the box's 57,954, −349. It is more restrictive, giving up 441 correct to avoid 46 wrong — a trade worth taking only if a wrong answer cost more than 10 silences.

The cone loses for both roles at every slope swept, and the reason is the one that sank every other rate-based idea in this study: the OpenSky archive carries positions forward between updates, so a *position* is trustworthy whenever it appears, while anything derived by dividing by elapsed time is not. A glideslope is a rate in disguise.

8.2 The bearing test

Version 5 proposed asking whether the trajectory *points at* the aerodrome: take the track's course near its endpoint, take the bearing from that endpoint to the candidate, and require the two to agree within some angle. It was the first signal found that could add answers the abstention refused, and it was proposed in four forms — **rescue** (answer anyway when aligned), **veto** (require alignment even when the gate accepts), **replace** (alignment instead of the gate) and **rerank** (choose *which* aerodrome by alignment).

All four are re-measured here against the current `endpoint` baseline.

Variant	ADEP coverage	ADEP accuracy	ADEP score	Δ	ADES coverage	ADES accuracy	ADES score	Δ
base	80.96%	98.15%	72,737	—	69.09%	95.64%	57,112	—
rescue: gate OR align<=0.1	81.07%	98.03%	72,560	−177	69.65%	95.53%	57,360	+248
rescue: gate OR align<=0.25	81.13%	97.98%	72,498	−239	69.84%	95.44%	57,348	+236
rescue: gate OR align<=1.0	81.48%	97.73%	72,221	−516	70.15%	95.25%	57,212	+100
rescue: gate OR align<=3.0	82.49%	96.91%	71,176	−1,561	70.71%	94.94%	57,057	−55
veto: gate AND align<=0.1	2.63%	93.27%	1,994	−70,743	11.64%	97.28%	10,169	−46,943
veto: gate AND align<=0.25	5.01%	96.12%	4,213	−68,524	13.61%	97.55%	11,994	−45,118
veto: gate AND align<=1.0	17.24%	98.48%	15,647	−57,090	16.59%	97.85%	14,765	−42,347
veto: gate AND align<=3.0	39.84%	99.04%	36,808	−35,929	22.67%	98.15%	20,373	−36,739
replace: align<=0.1 only	2.74%	89.93%	1,817	−70,920	12.20%	96.59%	10,417	−46,695
replace: align<=0.25 only	5.19%	93.51%	3,974	−68,763	14.36%	96.51%	12,230	−44,882
replace: align<=1.0 only	17.76%	96.52%	15,131	−57,606	17.65%	96.18%	14,865	−42,247
replace: align<=3.0 only	41.37%	96.52%	35,247	−37,490	24.30%	95.97%	20,318	−36,794
rerank by alignment, then gate	51.50%	86.63%	29,340	−43,397	35.49%	86.78%	20,370	−36,742
rerank, gate OR align<=0.1	52.72%	84.73%	27,168	−45,569	37.12%	84.68%	19,074	−38,038
rerank, gate OR align<=0.25	54.37%	82.36%	24,350	−48,387	38.47%	82.32%	17,190	−39,922

The best variant of any kind is `rescue: gate OR align<=0.1`, worth +248 on arrivals — and −177 on departures, which is the role `endpoint` is actually used for. Nothing here is worth adopting.

The three variants that let bearing *remove* or *choose* answers are not close: the best of them gives up −35,929 on departures. Alignment is far too blunt to name an aerodrome — at 40 NM a runway subtends a couple of degrees, but so does every other aerodrome behind it on the same radial, and a track on final approach points at all of them equally.

This supersedes version 5's recommendation. That version scored the bearing rescue against `endpoint` alone and never against `trend`, which was already better than both — the comparison that mattered was the one not made.

8.3 Bearing applied to `trend`

The bearing variants above all modify `endpoint`. `trend` is the algorithm that ships for arrivals, and it fails differently — it has no gate, so it almost always answers, and its errors are misnamings rather than silences. The variant that suits it is therefore a *rerank* or a *tie-break*, not a rescue, and its candidate set is already narrowed by the vote rule, so alignment is asked a much easier question than on the endpoint side.

Variant	ADES coverage	ADES accuracy	ADES score	Δ vs shipped
base: trend as shipped	69.76%	97.33%	61,045	—
rerank by alignment	69.76%	69.07%	4,786	−56,259
tie-break by alignment within 0.5 NM	69.76%	97.57%	61,510	+465
tie-break by alignment within 1 NM	69.76%	97.75%	61,867	+822
tie-break by alignment within 2 NM	69.76%	97.91%	62,185	+1,140
tie-break by alignment within 5 NM	69.76%	95.77%	57,922	−3,123
veto: abstain when align > 0.1 deg	3.99%	99.87%	3,779	−57,266
veto: abstain when align > 0.25 deg	7.74%	99.80%	7,319	−53,726
veto: abstain when align > 1 deg	12.47%	99.74%	11,765	−49,280
veto: abstain when align > 3 deg	14.89%	99.58%	13,984	−47,061
veto: abstain when align > 10 deg	20.39%	99.35%	19,009	−42,036

tie-break by alignment within 2 NM improves on the shipped configuration by +1,140. That is a real gain and it is not currently adopted – it arrived after the defaults were set, and adopting it would mean shipping a value this study has measured once, on one period. It is the first thing a follow-up should confirm.

8.4 The vertical measures

Whether a track is climbing or descending can be read from the broadcast vertical rate, from a two-point altitude slope, or by least squares over the whole window. The three disagree, and version 5 could not say which was wrong.

Window (min)	Freshness	Flights	Broadcast rate vs OLS	2-point slope vs OLS
5	1. <30% fresh	29,251	35.63%	36.52%
5	2. 30-60%	2,701	55.42%	58.61%
5	3. 60-90%	941	56.85%	62.17%
5	4. >=90% fresh	679	65.10%	51.55%
10	1. <30% fresh	18,597	46.87%	54.92%
10	2. 30-60%	13,239	57.78%	70.01%
10	3. 60-90%	1,447	58.53%	66.69%
10	4. >=90% fresh	822	64.36%	54.99%
20	1. <30% fresh	12,015	50.27%	57.76%
20	2. 30-60%	10,532	59.85%	62.40%
20	3. 60-90%	11,206	62.75%	78.13%
20	4. >=90% fresh	1,048	71.28%	60.31%

Across the sample the broadcast rate disagrees with the least-squares slope on 38.41% of cases and the two-point slope on 39.32%; 44.86% of broadcast rates are exactly zero, and 10.22% of samples are fresh. The disagreement is concentrated where position updates are stale, which is the same explanation that sank the approach cone: a position survives the archive's carry-forward, a rate does not.

8.5 The sampler: bin-based against fixed-phase

OPDI thins the 1 Hz feed to roughly 5 s. The rule this study inherited is fixed-phase — `event_time % 5 == 0`, which keeps a row only if one exists at the single second per window congruent to zero. The rule it now uses keeps the newest row in each 5 s bucket, and cannot drop a sample where the modulo rule would.

The bin-based rule is adopted, and **not because it measures better**. It is the sampler the thinning was always meant to be; the modulo rule survives only because the OpenSky archive is a complete 1 Hz grid with carried-forward positions, so a row almost always exists at the congruent second. That is a property of the archive, not a guarantee. If it ever changes, the modulo rule degrades badly and this one degrades gracefully.

What it costs is the question worth answering, and the answer is: nothing measurable.

Comparison	Role	Cells	Median Δscore	Median Δcoverage	Improved	Worsened
trend sweep	ADEP	371	+6	+0.0021 pp	231	134
trend sweep	ADES	371	+17	+0.0042 pp	310	57
endpoint grid	ADEP	126	-6	+0.0011 pp	41	84
endpoint grid	ADES	126	-12	-0.0137 pp	28	92
pipeline modes	ADEP	6	-10	+0.0000 pp	2	4
pipeline modes	ADES	6	+16	+0.0026 pp	3	3

Across every parameter cell of two complete runs, the largest median coverage difference in either direction is **0.0137 pp**. The trend side is consistently positive — 310 of 371 arrival cells improve, which is a real if tiny effect — and the endpoint side is consistently slightly negative. The shipped configuration takes departures from `endpoint` and arrivals from `trend`, so the two roughly cancel.

So the sampler is **neutral on accuracy and coverage**, and is adopted on correctness. It should not be described as an improvement.

> **i** Two earlier accounts of this were wrong, and how
>
> The decimation study measured **+0.26 pp** of arrival coverage for the bucket rule. That was one day, at one operating point, before the candidate ranking changed — and it does not reproduce. Measured across three days on the same rule it is −0.0137 pp.
> An end-to-end harness then reported the bucket rule *collapsing* arrival coverage from 80.96% to 27.65%. That was not the sampler either: its two arms had drifted onto different periods, comparing a three-day cache against a one-day one, and by then the arm labelled "modulo" was reading a table already rebuilt with the bucket rule. A third of the flights, a third of the coverage.
> Both are instructive in the same way. A sampler that changes 0.2% of rows cannot move coverage by fifty points or by a quarter of a point; when a measurement says otherwise, the measurement is what to check.

9 What the pipeline actually produces

Every number so far comes from a sweep harness, not from `src/opdi`. The trend sweep re-applies the vote rule to a cached vote table and the endpoint sweeps filter and re-rank a cached candidate table — which is what makes hundreds of parameter cells affordable, and also what makes them unfit, on their own, to justify changing a production default.

So the recommended settings were run through `FlightListProcessor.process_dai` itself and scored the same way. Three differences between harness and pipeline are known in advance and are not bugs:

1. the pipeline selects among candidate aerodromes by **H3 ring count** (`distance_from_center`, an integer stepping about 5.2 km at resolution 7), while the harness uses exact haversine distance;
2. the pipeline filters zones by radius **band**, the harness by exact radius;
3. the harness smooths barometric altitude across the flight-level boundary and the pipeline does not.

The gap those produce is the fidelity of every tuned number on this page.

Run	Role	Mode	Coverage	Accuracy	Correct	Wrong	Score
legacy	ADEP	trend	68.72%	98.83%	64,600	768	63,064
legacy	ADES	trend	68.06%	97.87%	63,358	1,379	60,600
trend	ADEP	trend	78.06%	98.50%	73,127	1,116	70,895
trend	ADES	trend	72.18%	94.39%	64,802	3,849	57,104
endpoint	ADEP	endpoint	80.96%	98.15%	75,581	1,422	72,737
endpoint	ADES	endpoint	69.09%	95.64%	62,848	2,868	57,112
nearest	ADEP	nearest	91.68%	87.78%	76,545	10,660	55,225
nearest	ADES	nearest	88.55%	79.44%	66,904	17,318	32,268
combined	ADEP	endpoint	80.90%	98.22%	75,579	1,371	72,837
combined	ADES	trend	68.27%	97.95%	63,604	1,330	60,944
recommended	ADEP	endpoint	80.90%	98.22%	75,579	1,371	72,837
recommended	ADES	trend	68.27%	97.95%	63,604	1,330	60,944

Harness against pipeline, at the same parameters:

Role	Method	Harness score	Pipeline score	Residual	Relative
ADEP	endpoint	72,737	72,737	+0	+0.00%
ADES	trend	61,069	60,944	−125	-0.20%

The two rows could hardly be more different, and the difference is the single most important result on this page.

`endpoint` **reproduces exactly**. That is not luck: the endpoint sweep filters the same cached candidate table the pipeline reads, applying the same rule, so harness and pipeline are the same computation and agree to the flight.

`trend` **does not**. The harness ranks candidate aerodromes by exact haversine distance. The pipeline first keeps only the candidates at the minimum H3 *ring count* — an integer stepping about 5.2 km at resolution 7 — and applies the haversine tie-break, and the scheduled-service penalty, only within that surviving set. An aerodrome one ring further out is gone before the penalty can promote it. So a penalty tuned against exact distance does something weaker in production, and the harness's trend optimum is optimistic by the margin shown.

9.1 What running the pipeline caught

The first pipeline path run produced a result that no amount of sweeping could have: the scheduled-service penalty step scored **exactly** what the step before it did, to the flight. An identical score is not a weak effect, it is no effect, and a parameter with no effect is a parameter that is not reaching the code.

It was not. The penalty was added to the ranking in `_compute_flight_table`, but the projection a few lines above it dropped the `apt_scheduled` column, so the ranking looked the column up, failed to find it, and fell back to raw distance. A penalty of ten nautical miles and a penalty of zero produced byte-identical flight lists.

Two things made this visible rather than invisible. The pipeline run put the parameter in front of real data, where a sweep harness — which reads a *different* table that does carry the column — could not. And the fallback printed a note saying the column was missing instead of quietly ranking on distance, because a missing column and a zero penalty look identical in the output. The numbers below are from after the fix.

9.2 The trend tuning re-walked inside the pipeline

The path below runs through `process_dai` itself: production constants, then one tuned value adopted per step. It was run twice — once with the pipeline's original **H3 ring-count** selection, and once with **exact-distance** ranking. The two disagree in sign, which is the whole finding.

Step	Ring: score	Ring: gain	Exact: score	Exact: gain	Coverage	Accuracy
production constants	60,603	—	60,255	—	68.06%	97.69%
+ scheduled-service penalty	60,855	+252	61,113	+858	68.06%	98.13%
+ flight-level cap	59,136	−1,719	61,773	+660	69.92%	97.63%

Step	Ring: score	Ring: gain	Exact: score	Exact: gain	Coverage	Accuracy
+ vote margin	59,295	+159	61,941	+168	70.17%	97.60%
+ radius	59,454	+159	62,004	+63	69.84%	97.78%

Under ring selection the tuning is worth −1,149; under exact distance it is worth +1,749. The flight-level cap is the step that flips hardest — −1,719 against +660 — and it is the step that admits samples from higher up, where more aerodromes share a ring and the ring rule is least able to separate them.

So the sweep harness was never over-estimating. It was measuring a *better algorithm* — one that measures every candidate before choosing — and the pipeline has now adopted it. `trend_rank_by` defaults to `"haversine"`; `DetectionConfig.legacy()` still returns `"ring"`, so every published flight list stays reproducible.

9.3 Production's own optimum

The path adopts the values the *harness* preferred. Whether those are also the **pipeline's** optima is a separate question, and it is answered by sweeping the flight-level cap and radius through `process_dai` directly.

Score by flight-level cap (rows) and radius (columns), through the pipeline, at vote margin 2 and penalty 10 NM. Departures are shown for completeness – they are served by `endpoint`, so that half is a check rather than a recommendation.

FL cap	ADES 20 NM	ADES 30 NM	ADEP 20 NM	ADEP 30 NM
40	61,554	61,329	64,408	64,143
60	62,004	61,941	67,506	67,265
80	61,146	61,373	69,121	68,891
100	59,007	59,548	70,374	70,195
120	57,154	57,667	70,927	70,882

Arrivals peak at **FL60 / 20 NM** and fall away on both sides — an interior optimum, not an edge of the grid. Departures climb monotonically to **FL120**, but departures are served by `endpoint`, so that column is a check rather than a recommendation. Both roles prefer the tighter radius at every flight level.

These are the values that ship. Both had been rejected earlier in this study on ring-ranked evidence, and both are reinstated here on the pipeline's own measurement.

One caveat, stated rather than buried: the grid holds the **vote margin** fixed at 2 while varying the cap and radius. That value therefore rests on the sweep harness alone, and is the weakest-supported of the shipped parameters.

10 Why each value, and not its neighbour

A recommendation is only as good as the evidence that picked it, so every value the sweeps select appears below with the score at that value, the score at the adjacent grid points, and the score at whatever production uses today. The last column is the one that matters most for trusting the number: a value on a broad **plateau** is a safe recommendation even from a single sample, because its neighbours score nearly the same and a little noise cannot move it far. A value on a sharp isolated **peak** may be partly the sample's noise, and is flagged as such however good its score.

> **!** These are the sweeps' choices, not the shipped ones
>
> The `Chosen` column below is the **harness argmax**, which is what this section exists to scrutinise. For `trend`, Section 9 is what settles whether each value survives in production — and under exact-distance ranking they do, which is why the harness numbers below and the shipped values in Section 15 now agree. They are still kept apart, because for a period they did *not* agree, and a sweep preference is only ever evidence until the pipeline confirms it.

Role	Parameter	Production	Sweep argmax	Score at production	Score at argmax	Neighbour below	Neighbour above	Δ vs production	Shape
ADES	trend FL cap	40	60	60,736	61,069	60,736	60,269	+333	peak
ADES	trend vote margin	4	2	60,865	61,069	61,026	60,865	+204	plateau
ADES	trend radius (NM)	30	20	61,057	61,069	59,771	61,057	+12	plateau
ADEP	endpoint radius (NM)	40	30	72,522	72,737	72,586	72,522	+215	plateau
ADEP	endpoint height (ft)	15000	15000	72,737	72,737	71,571	72,355	+0	peak

The penalty is treated separately because it is not a geometric parameter: it re-ranks candidates without changing how many are answered, so its sweep holds the geometry fixed and appears in Section 6.

10.1 Which change is actually doing the work

The table above varies one parameter at a time around the chosen setting, which shows whether each value is defensible but not how much each contributes to the total gain. This walks the whole distance instead: from the production constants to the recommended setting, changing one parameter per step.

Step	Score	Gain from this step	Share of total gain
production constants (FL40, margin 4, 30 NM, no penalty)	59,321	—	—
+ scheduled-service penalty of 10 NM	60,227	+906	52%
+ flight-level cap FL60	60,856	+629	36%
+ vote margin 2	61,057	+201	11%
+ radius 20 NM	61,069	+12	1%

The gain is not spread evenly. **scheduled-service penalty of 10 NM** accounts for 52% of it on its own. That is worth stating plainly: the single most valuable change to arrival detection is not a tuned threshold at all, but applying to `trend` the scheduled-service tie-break that `endpoint` already had and `trend` never did.

By contrast the radius step is worth 12 flights out of 61,069 here — inside what one three-day sample can resolve, and on this evidence alone it would not justify moving a published constant. It is adopted anyway, on stronger evidence gathered later: the pipeline grid in Section 9 prefers the tighter radius at *every* flight level for *both* roles, which is a consistent effect across ten cells rather than a single argmax.

11 The verdict

Configuration	ADEP	ADES	ADEP coverage	ADEP accuracy	ADEP score	ADES coverage	ADES accuracy	ADES score	Total
SHIPPED endpoint + exact-distance trend	endpoint	trend	80.90%	98.22%	72,837	68.27%	97.95%	60,944	133,781
control: production geometry + penalty, ring-ranked	endpoint	trend	80.90%	98.22%	72,837	68.27%	97.95%	60,944	133,781
endpoint, both roles	endpoint	endpoint	80.96%	98.15%	72,737	69.09%	95.64%	57,112	129,849
trend, departure parameters both roles	trend	trend	78.06%	98.50%	70,895	72.18%	94.39%	57,104	127,999
legacy (production today)	trend	trend	68.72%	98.83%	63,064	68.06%	97.87%	60,600	123,664
nearest (traffic library)	nearest	nearest	91.68%	87.78%	55,225	88.55%	79.44%	32,268	87,493

The recommended configuration is the **combined** row: `endpoint` for departures, exact-distance `trend` for arrivals. It scores **133,781 against production's 123,664, a gain of +10,117** on a three-day sample of 95,116 flights, and every figure in that table is a real pipeline run scored against Network Manager ground truth.

It is the shipped configuration in the strict sense: its trend cap, radius, vote margin, penalty and ranking rule, and its endpoint radius, height and penalty, were checked field by field against `DetectionConfig()`. The row is what production produces, not an approximation of it.

The `recommended` row is a **control, and a mislabelled one**. It was built by starting from `DetectionConfig.legacy()` and overriding the fields the author remembered to override – and `trend_rank_by` was not among them, so it kept ring-count selection and scored +10,117 against production while every properly-ranked configuration gained. It is left in the table because it is the cleanest demonstration on this page that the ranking rule matters more than the thresholds: identical geometry, one different selection rule, +0 of score.

11.1 What is being recommended, and why

Departures — `endpoint`, 30 NM, 15,000 ft, 10 NM scheduled-service penalty. Score 72,837 against production's 63,064, a gain of +9,773. The clearest result in the study: an interior optimum in both parameters, better than `trend` at every exchange rate tried, and the pipeline reproduces the sweep exactly – the endpoint sweep filters the same cached candidate table the pipeline reads, so the tuned figure *is* the production figure.

Arrivals — `trend` at FL60, vote margin 2, 20 NM, 10 NM penalty, ranked on exact distance. Measured as an algorithm, on a trend-only flight list, that is 62,004 against production's 60,600 – a gain of +1,404, won by converting 1,594 silences into correct answers for 95 extra wrong ones.

Inside the combined flight list the arrival half reads 60,944, +344 against production rather than the +1,404 just quoted. The difference is not the algorithm. Taking departures from `endpoint` adds tracks `trend` never produced, and where two share an aircraft, callsign and date, the scoring alignment awards the flight to whichever starts closest to the actual off-block time – sometimes now a departure-only track carrying no arrival. Accuracy *rises* in the merged table (97.95% against 97.87%) while coverage falls, which is the signature of lost pairings rather than wrong names.

One weak link, stated plainly. The vote margin of 2 comes from the sweep harness; the pipeline grid held it fixed while varying the cap and radius. Every other shipped value was measured through `process_dai`. If one parameter in this recommendation is going to fail on a fresh sample, that is the one.

11.2 The `traffic` library's rule, for reference

`nearest` — the unconditional rule the `traffic` library uses — answers far more often than anything else here (91.68% of departures, 88.55% of arrivals) and is right far less often (87.78% and 79.44%). Scored, it is last by a wide margin: 87,493 against the recommendation's 133,781.

That is not a criticism of `traffic`, which is solving a different problem: it names the nearest aerodrome and leaves the judgement to the caller. It is, however, the measurement that justifies OPDI carrying its own rule rather than adopting that one.

12 By aerodrome class

The candidate set is large and medium aerodromes only, so a flight whose true aerodrome is a small airfield or a heliport cannot be named correctly by any method on this page. This is the size of that population, and of everything else.

Departures

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,661	331	71,167	70,588	88.23%	99.19%
out of area	7,892	1	920	630	11.66%	68.48%

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
medium_airport	6,069	352	4,703	4,361	77.49%	92.73%
small_airport	361	76	132	0	36.57%	0.00%
heliport	133	65	28	0	21.05%	0.00%

Arrivals

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,683	331	61,050	60,260	75.67%	98.71%
out of area	7,964	1	25	0	0.31%	0.00%
medium_airport	5,983	346	3,764	3,344	62.91%	88.84%
small_airport	352	70	84	0	23.86%	0.00%
heliport	134	58	11	0	8.21%	0.00%

Outside the candidate set — 3 classes, 16,836 flights across both roles — the method is working against a reference it was never given. Those flights are counted in the denominator throughout this page, so every coverage figure here already carries their cost.

13 The busiest 100 aerodromes

Movement counts are what OPDI is for, so the last check is per aerodrome rather than per flight: for each of the hundred busiest in the reference, how many movements the recommended methodology finds, against how many the Network Manager data says there were, and how the algorithm production runs today compares on the same aerodromes.

`Count ratio` is movements found over movements in truth — the number a capacity or delay analysis would be distorted by. `Recall` is the share of that aerodrome's true movements correctly attributed to it.

13.1 Departures

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
1	LTFM	2,282	2,209	2,202	0.968	96.49%	67.18%	+29.32 pp
2	EHAM	2,148	2,143	2,143	1.005	99.77%	99.81%	-0.05 pp
3	EDDF	2,094	2,082	2,082	0.995	99.43%	96.94%	+2.48 pp
4	LFPG	2,084	2,065	2,061	0.994	98.90%	99.04%	-0.14 pp
5	EGLL	2,001	1,989	1,988	1.011	99.35%	99.30%	+0.05 pp
6	LEMD	1,773	1,760	1,759	0.992	99.21%	94.36%	+4.85 pp
7	LEBL	1,591	1,578	1,577	0.991	99.12%	99.43%	-0.31 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
8	EDDM	1,573	1,562	1,562	0.995	99.30%	99.17%	+0.13 pp
9	LIRF	1,449	1,442	1,442	0.995	99.52%	99.72%	-0.21 pp
10	LTAI	1,435	1,371	1,370	0.960	95.47%	61.46%	+34.01 pp
11	LEPA	1,417	1,397	1,397	0.987	98.59%	92.52%	+6.07 pp
12	LGAV	1,306	1,239	1,239	0.956	94.87%	23.20%	+71.67 pp
13	EGKK	1,223	1,214	1,214	0.994	99.26%	99.67%	-0.41 pp
14	EIDW	1,185	1,164	1,163	0.981	98.14%	98.23%	-0.08 pp
15	LSZH	1,178	1,170	1,162	0.989	98.64%	98.22%	+0.42 pp
16	LOWW	1,156	1,138	1,137	0.990	98.36%	98.88%	-0.52 pp
17	EKCH	1,140	1,133	1,133	0.995	99.39%	97.98%	+1.40 pp
18	LTFJ	1,137	1,101	1,100	0.979	96.75%	93.58%	+3.17 pp
19	LPPT	1,010	1,001	1,001	0.991	99.11%	80.30%	+18.81 pp
20	LFPO	992	982	982	0.997	98.99%	99.29%	-0.30 pp
21	LIMC	973	960	959	0.987	98.56%	99.28%	-0.72 pp
22	ENGM	932	924	924	0.997	99.14%	91.95%	+7.19 pp
23	EGCC	926	909	908	0.981	98.06%	99.35%	-1.30 pp
24	EGSS	885	881	881	0.995	99.55%	99.44%	+0.11 pp
25	EPWA	875	869	868	0.997	99.20%	99.20%	+0.00 pp
26	EBBR	872	868	868	1.008	99.54%	99.54%	+0.00 pp
27	EDDB	870	851	849	0.992	97.59%	97.47%	+0.11 pp
28	LEMG	850	847	847	0.999	99.65%	99.53%	+0.12 pp
29	ESSA	820	791	787	0.965	95.98%	96.34%	-0.37 pp
30	EDDL	758	745	743	0.987	98.02%	99.08%	-1.06 pp
31	LFMN	707	699	693	0.982	98.02%	97.74%	+0.28 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
32	LSGG	680	677	677	1.012	99.56%	67.94%	+31.62 pp
33	LKPR	667	656	655	0.988	98.20%	98.65%	-0.45 pp
34	EFHK	659	653	653	1.003	99.09%	99.24%	-0.15 pp
35	LEAL	599	597	597	0.997	99.67%	99.83%	-0.17 pp
36	LHBP	578	553	551	0.955	95.33%	97.92%	-2.60 pp
37	EGPH	577	570	569	0.986	98.61%	98.96%	-0.35 pp
38	EGGW	550	549	549	1.002	99.82%	98.00%	+1.82 pp
39	EDDH	538	533	530	1.002	98.51%	98.33%	+0.19 pp
40	GCLP	520	23	0	0.000	0.00%	0.00%	+0.00 pp
41	EDDK	515	501	501	0.975	97.28%	96.89%	+0.39 pp
42	LLBG	513	499	499	0.992	97.27%	97.08%	+0.19 pp
43	LPPR	513	511	508	0.990	99.03%	99.03%	+0.00 pp
44	LROP	504	491	491	0.974	97.42%	98.21%	-0.79 pp
45	LEIB	495	490	490	0.990	98.99%	89.29%	+9.70 pp
46	EGBB	474	468	468	0.987	98.73%	98.73%	+0.00 pp
47	LIML	458	451	451	0.987	98.47%	98.47%	+0.00 pp
48	LIME	453	450	450	0.993	99.34%	98.68%	+0.66 pp
49	LIRN	449	380	334	0.751	74.39%	0.00%	+74.39 pp
50	LFML	444	441	440	0.991	99.10%	99.10%	+0.00 pp
51	EDDS	440	421	415	0.950	94.32%	94.09%	+0.23 pp
52	LGIR	424	6	0	0.000	0.00%	0.00%	+0.00 pp
53	LIPZ	412	408	408	0.993	99.03%	98.79%	+0.24 pp
54	LFLL	411	402	402	1.015	97.81%	98.78%	-0.97 pp
55	LPFR	405	402	400	0.988	98.77%	98.77%	+0.00 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
56	LYBE	394	388	388	0.987	98.48%	98.22%	+0.25 pp
57	LICC	390	384	380	0.974	97.44%	0.00%	+97.44 pp
58	LTAC	381	7	0	0.000	0.00%	0.00%	+0.00 pp
59	LIPE	380	377	377	0.992	99.21%	98.42%	+0.79 pp
60	LCLK	368	353	352	0.959	95.65%	86.41%	+9.24 pp
61	GMMN	360	321	316	1.072	87.78%	62.22%	+25.56 pp
62	EPKK	360	349	347	0.967	96.39%	98.61%	-2.22 pp
63	EGGD	359	356	355	0.989	98.89%	99.44%	-0.56 pp
64	LEVC	355	342	341	0.961	96.06%	90.42%	+5.63 pp
65	LFSB	354	351	349	0.994	98.59%	98.31%	+0.28 pp
66	ENBR	349	343	343	1.026	98.28%	96.85%	+1.43 pp
67	LTBJ	346	327	279	0.806	80.64%	1.45%	+79.19 pp
68	EGPF	344	325	323	0.951	93.90%	93.60%	+0.29 pp
69	LGRP	340	327	326	0.976	95.88%	93.82%	+2.06 pp
70	LMML	326	325	325	1.000	99.69%	98.77%	+0.92 pp
71	GCTS	321	5	0	0.019	0.00%	0.00%	+0.00 pp
72	GCXO	319	306	306	1.141	95.92%	21.32%	+74.61 pp
73	LATI	316	7	0	0.000	0.00%	0.00%	+0.00 pp
74	LICJ	310	286	286	0.926	92.26%	54.19%	+38.06 pp
75	EBCI	304	304	304	1.000	100.00%	100.00%	+0.00 pp
76	BIKF	297	296	294	1.084	98.99%	98.32%	+0.67 pp
77	ELLX	294	291	291	0.990	98.98%	98.98%	+0.00 pp
78	LFBO	293	289	289	0.993	98.63%	98.98%	-0.34 pp
79	LGTS	279	272	271	0.971	97.13%	68.82%	+28.32 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
80	GMMX	278	93	86	0.317	30.94%	0.36%	+30.58 pp
81	LEZL	275	274	273	0.993	99.27%	99.64%	-0.36 pp
82	LTBS	273	219	219	0.802	80.22%	0.00%	+80.22 pp
83	EVRA	273	271	270	0.993	98.90%	98.17%	+0.73 pp
84	LBSF	272	162	162	0.596	59.56%	0.00%	+59.56 pp
85	GCRR	269	1	0	0.000	0.00%	0.00%	+0.00 pp
86	EGNX	267	263	263	0.985	98.50%	98.88%	-0.37 pp
87	EPKT	259	251	249	0.961	96.14%	93.44%	+2.70 pp
88	HEGN	257	0	0	0.000	0.00%	0.00%	+0.00 pp
89	UGTB	257	165	159	0.774	61.87%	0.00%	+61.87 pp
90	LFRS	256	253	253	0.992	98.83%	78.91%	+19.92 pp
91	EDDP	256	255	255	1.004	99.61%	98.05%	+1.56 pp
92	EPGD	248	55	40	0.161	16.13%	6.45%	+9.68 pp
93	LGKR	244	209	207	0.869	84.84%	0.00%	+84.84 pp
94	HECA	244	62	42	0.172	17.21%	6.97%	+10.25 pp
95	EGAA	239	239	239	1.004	100.00%	76.15%	+23.85 pp
96	ENZV	236	163	160	0.737	67.80%	30.93%	+36.86 pp
97	LEBB	234	183	172	0.744	73.50%	0.00%	+73.50 pp
98	LFPB	230	227	227	1.000	98.70%	95.65%	+3.04 pp
99	LIBD	228	1	0	0.000	0.00%	0.00%	+0.00 pp
100	LIEO	228	201	195	0.855	85.53%	7.89%	+77.63 pp

13.2 Arrivals

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
1	LTFM	2,258	1,817	1,770	0.784	78.39%	71.26%	+7.13 pp
2	EHAM	2,136	1,879	1,873	0.877	87.69%	90.17%	-2.48 pp
3	EDDF	2,091	1,775	1,770	0.847	84.65%	87.61%	-2.97 pp
4	LFPG	2,066	1,715	1,714	0.852	82.96%	85.82%	-2.86 pp
5	EGLL	1,986	1,581	1,447	0.794	72.86%	85.45%	-12.59 pp
6	LEMD	1,786	1,534	1,532	0.861	85.78%	90.26%	-4.48 pp
7	LEBL	1,597	1,498	1,495	0.937	93.61%	95.62%	-2.00 pp
8	EDDM	1,571	1,469	1,466	0.934	93.32%	94.27%	-0.95 pp
9	LTAI	1,437	940	940	0.656	65.41%	62.21%	+3.20 pp
10	LEPA	1,436	1,415	1,414	0.986	98.47%	98.05%	+0.42 pp
11	LIRF	1,433	1,255	1,253	0.880	87.44%	90.16%	-2.72 pp
12	LGAV	1,317	632	626	0.477	47.53%	12.45%	+35.08 pp
13	EGKK	1,220	1,101	1,100	0.902	90.16%	92.54%	-2.38 pp
14	EIDW	1,188	1,084	1,083	0.914	91.16%	93.27%	-2.10 pp
15	LSZH	1,160	1,052	1,050	0.914	90.52%	91.98%	-1.47 pp
16	EKCH	1,152	1,058	1,044	0.906	90.62%	91.93%	-1.30 pp
17	LOWW	1,145	1,047	1,043	0.914	91.09%	93.54%	-2.45 pp
18	LTFJ	1,138	994	984	0.878	86.47%	87.61%	-1.14 pp
19	LPPT	1,001	880	876	0.876	87.51%	90.51%	-3.00 pp
20	LFPO	991	940	940	0.950	94.85%	95.76%	-0.91 pp
21	LIMC	967	856	852	0.883	88.11%	91.62%	-3.52 pp
22	ENGM	940	889	889	0.952	94.57%	95.21%	-0.64 pp
23	EGCC	924	820	819	0.887	88.64%	90.91%	-2.27 pp
24	EGSS	888	835	830	0.938	93.47%	94.71%	-1.24 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
25	EPWA	880	782	751	0.853	85.34%	88.64%	-3.30 pp
26	EBBR	871	767	764	0.885	87.72%	89.90%	-2.18 pp
27	LEMG	861	834	834	0.969	96.86%	97.91%	-1.05 pp
28	ESSA	841	727	724	0.870	86.09%	86.21%	-0.12 pp
29	EDDB	840	789	784	0.937	93.33%	94.64%	-1.31 pp
30	EDDL	751	716	715	0.960	95.21%	97.20%	-2.00 pp
31	LFMN	710	684	680	1.010	95.77%	96.48%	-0.70 pp
32	LSGG	670	637	634	0.954	94.63%	96.72%	-2.09 pp
33	EFHK	666	574	573	0.865	86.04%	86.64%	-0.60 pp
34	LKPR	657	604	602	0.919	91.63%	93.76%	-2.13 pp
35	LEAL	599	589	588	0.982	98.16%	98.83%	-0.67 pp
36	EGPH	584	541	541	0.933	92.64%	94.69%	-2.05 pp
37	LHBP	577	518	515	0.898	89.25%	93.59%	-4.33 pp
38	EGGW	552	529	521	0.947	94.38%	95.83%	-1.45 pp
39	EDDH	532	509	507	0.970	95.30%	97.74%	-2.44 pp
40	GCLP	521	0	0	0.000	0.00%	0.00%	+0.00 pp
41	LLBG	516	432	426	0.828	82.56%	88.37%	-5.81 pp
42	LPPR	512	479	478	0.934	93.36%	93.75%	-0.39 pp
43	LEIB	509	498	498	0.980	97.84%	97.64%	+0.20 pp
44	EDDK	509	473	470	0.923	92.34%	95.09%	-2.75 pp
45	LROP	509	486	484	0.953	95.09%	97.25%	-2.16 pp
46	EGBB	466	419	418	0.897	89.70%	92.70%	-3.00 pp
47	LIML	465	457	455	0.987	97.85%	97.63%	+0.22 pp
48	LIME	460	447	444	0.965	96.52%	97.83%	-1.30 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
49	LIRN	447	2	0	0.000	0.00%	0.00%	+0.00 pp
50	LFML	445	439	437	0.991	98.20%	99.10%	-0.90 pp
51	EDDS	433	409	409	0.958	94.46%	95.61%	-1.15 pp
52	LGIR	430	0	0	0.000	0.00%	0.00%	+0.00 pp
53	LFLL	412	398	396	0.995	96.12%	97.33%	-1.21 pp
54	LIPZ	409	388	386	0.946	94.38%	95.35%	-0.98 pp
55	LPFR	407	403	402	0.993	98.77%	98.77%	+0.00 pp
56	LYBE	395	364	364	0.922	92.15%	94.68%	-2.53 pp
57	LICC	391	345	326	0.834	83.38%	5.88%	+77.49 pp
58	LIPE	376	370	369	0.981	98.14%	98.94%	-0.80 pp
59	LTAC	374	0	0	0.000	0.00%	0.00%	+0.00 pp
60	LCLK	367	348	346	0.954	94.28%	97.28%	-3.00 pp
61	EGGD	360	334	334	0.933	92.78%	95.28%	-2.50 pp
62	LEVC	357	348	348	0.978	97.48%	98.04%	-0.56 pp
63	GMMN	357	266	266	0.754	74.51%	77.87%	-3.36 pp
64	ENBR	357	341	329	0.944	92.16%	92.16%	+0.00 pp
65	EPKK	356	342	333	0.938	93.54%	94.66%	-1.12 pp
66	LFSB	352	340	339	0.966	96.31%	97.73%	-1.42 pp
67	LGRP	346	316	315	0.913	91.04%	91.04%	+0.00 pp
68	EGPF	346	301	300	0.876	86.71%	88.44%	-1.73 pp
69	LTBJ	341	14	14	0.041	4.11%	0.00%	+4.11 pp
70	LMML	326	318	316	0.969	96.93%	97.85%	-0.92 pp
71	LATI	323	0	0	0.000	0.00%	0.00%	+0.00 pp
72	GCTS	323	0	0	0.000	0.00%	0.00%	+0.00 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
73	GCXO	320	2	2	0.006	0.62%	0.00%	+0.62 pp
74	LICJ	314	138	138	0.443	43.95%	38.22%	+5.73 pp
75	EBCI	303	297	296	0.977	97.69%	98.68%	-0.99 pp
76	BIKF	296	90	88	0.301	29.73%	33.11%	-3.38 pp
77	ELLX	294	271	271	0.925	92.18%	96.60%	-4.42 pp
78	LFBO	290	287	285	0.983	98.28%	98.97%	-0.69 pp
79	LGTS	281	265	265	0.943	94.31%	91.81%	+2.49 pp
80	LTBS	280	8	8	0.029	2.86%	0.36%	+2.50 pp
81	GMMX	279	14	13	0.047	4.66%	0.00%	+4.66 pp
82	EVRA	278	227	226	0.817	81.29%	82.37%	-1.08 pp
83	LEZL	276	264	263	0.957	95.29%	99.28%	-3.99 pp
84	GCRR	269	0	0	0.000	0.00%	0.00%	+0.00 pp
85	LBSF	267	2	1	0.004	0.37%	0.00%	+0.37 pp
86	EDDP	261	236	235	0.904	90.04%	93.49%	-3.45 pp
87	HEGN	258	0	0	0.000	0.00%	0.00%	+0.00 pp
88	LFRS	257	251	250	0.977	97.28%	97.28%	+0.00 pp
89	UGTB	257	48	47	0.187	18.29%	19.46%	-1.17 pp
90	EPKT	257	242	239	0.930	93.00%	93.39%	-0.39 pp
91	EGNX	255	231	231	0.906	90.59%	93.33%	-2.75 pp
92	LIEO	252	110	109	0.433	43.25%	21.03%	+22.22 pp
93	LGKR	247	29	28	0.113	11.34%	0.00%	+11.34 pp
94	EGAA	246	234	149	0.606	60.57%	89.84%	-29.27 pp
95	EPGD	244	22	1	0.012	0.41%	0.41%	+0.00 pp
96	ENZV	236	70	67	0.301	28.39%	27.97%	+0.42 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
97	LEBB	235	0	0	0.000	0.00%	0.00%	+0.00 pp
98	LTFE	232	199	199	0.879	85.78%	78.02%	+7.76 pp
99	HECA	231	75	61	0.264	26.41%	2.16%	+24.24 pp
100	LIBD	229	1	0	0.000	0.00%	0.00%	+0.00 pp

13.3 Across the hundred

Role	Movements in truth	Recall, shipped	Recall, production	Aerodromes improved	Median count ratio
Departures	65,382	92.37%	83.41%	59 of 100	0.988
Arrivals	65,360	79.59%	79.74%	19 of 100	0.903

A count ratio below one means the aerodrome is under-counted — the usual case, since every abstention is a movement not attributed. Above one means movements are being attributed to an aerodrome that did not see them, which is the more damaging error and the one the scoring penalises at rate k.

The departure row is the result: recall rises by nearly nine points and the median aerodrome is within about one percent of its true movement count.

The arrival row must be read with Section 11 in hand. Recall reads 79.59% against production's 79.74%, improving on 19 of 100 — which looks like a mild regression and is not one. Measured as an algorithm on a trend-only flight list, arrivals gain; the shortfall here comes from the merged list also containing departure-only tracks, which this table pairs with reference flights by start time, so some arrivals are counted against a track that carries none. The measurement moved; the data did not. It is left in the table rather than corrected away, because a benchmark that quietly adjusts its own alignment to flatter a result is worth less than one that shows where it is fragile.

14 Does any of it hold on a second period?

Everything above is one three-day sample. An argmax over hundreds of cells on a single sample always flatters itself: part of the winning cell's margin is that sample's noise, and noise does not reproduce. The only way to find out which part is real is to measure a different period with the same code.

The second period is **5–7 June 2024**, chosen because the Network Manager reference for that month is committed to the repository and the matching tracks already exist from earlier versions of this study.

The 2024 sample carries 92,799 ground-truth flights against 2025's 95,116.

Role	Parameter	2025 optimum	2024 optimum	Agrees
ADEP	trend FL cap	120	100	**no**
ADEP	trend vote margin	0	0	yes
ADEP	trend radius (NM)	20	20	yes
ADES	trend FL cap	60	80	**no**
ADES	trend vote margin	2	2	yes
ADES	trend radius (NM)	20	30	**no**

3 of 6 parameter optima agree across the two periods.

On the arrival side, where this study rejected the sweep's preferences, the 2024 sweep prefers FL80 against 2025's FL60 — and its own gain over the production constants is +5,063, against +1,748 on 2025. Both are harness figures, and Section 9 showed harness gains of this size on `trend` do not survive the pipeline's ranking. A parameter that is neither stable across periods nor reproducible in production is not one to ship, which is what the defaults reflect.

For the endpoint rule, which is what departures actually adopt:

Role	2025 radius / height	2024 radius / height	Agrees
ADEP	30 NM / 15,000 ft	30 NM / 15,000 ft	yes
ADES	30 NM / 10,000 ft	30 NM / 10,000 ft	yes

15 What ships

The pipeline's `DetectionConfig` defaults are now the values this study measured. Six changed. The ordering below is deliberate: the ranking rule comes first, because without it none of the rest holds.

Parameter	Production	Now ships	Evidence
`trend_rank_by`	`ring`	**`haversine`**	The enabling change. Ring-count selection discarded candidates before measuring them; every tuned trend value fails under it and gains under exact distance.
`trend_max_fl`	40	**60**	Pipeline grid over FL 40–120: arrivals peak at FL60 and fall on both sides.
`trend_radius_nm`	30	**20**	Preferred at every flight level in the grid, both roles.

Parameter	Production	Now ships	Evidence
`trend_vote_margin`	4	**2**	Sweep harness only — the grid held it fixed. The weakest-supported value here.
`trend_sched_penalty_nm`	0	**10**	A tie-break `trend` never had. Improves 9 of the hundred busiest aerodromes, worsens none.
`endpoint_radius_nm`	40	**30**	Interior optimum; pipeline reproduces the sweep exactly.
`endpoint_height_ft`	15,000	15,000	Confirmed where it already was.
`endpoint_sched_penalty_nm`	10	10	Unchanged.

Three of these — the cap, the radius and the vote margin — were **rejected earlier in this study and reinstated**. The rejection was measured under ring-count selection, where raising the cap genuinely did lose ground. The lesson is not that the first measurement was careless but that it answered a narrower question than it appeared to: *given this selection rule*, the tuning fails. Changing the rule changed the answer.

Every OPDI flight list published before this change was built with the production column, and `DetectionConfig.legacy()` still returns exactly those values — including `trend_rank_by = "ring"` — so anything already released stays reproducible by name. That is the same principle as the frozen `version` strings on published events. A regression test pins the preset, a second pins the shipped values, and a third asserts the two differ, so the preset cannot quietly become vacuous.

⚠ Re-running a past month will not reproduce it

These defaults change ADEP and ADES for any month re-processed with them. That is intended, and it is why the old behaviour remains reachable by name rather than being deleted. A new algorithm gets a new `version`; the published one is never mutated.

16 Provenance

Each row is one file this page reads: the job that produced it, the commit it was produced at, and when. Every figure on this page is regenerated by `benchmarks/regenerate_v6.py`, which the page runs before drawing anything, so a row marked **unverified** would mean a file staged outside that chain and traceable to no command.

File	Job	Commit	Produced
bearing_whole_sample_v6.csv	bearing_whole_sample.py	d6abb5d	2026-08-10T22:48
decimation_v6.csv	decimation_end_to_end.py	9599034	2026-08-10T23:43

File	Job	Commit	Produced
merge_agreement_v6.csv	merge_diagnosis.py	d3130e7 (dirty)	2026-08-11T00:20
merge_shapes_v6.csv	merge_diagnosis.py	d3130e7 (dirty)	2026-08-11T00:20
mode_comparison_v6.csv	flight_list_v6.py	0399ea3	2026-08-10T23:06
per_airport_path_v6.csv	flight_list_v6.py	532610d	2026-08-11T00:10
per_airport_v6.csv	flight_list_v6.py	0399ea3	2026-08-10T23:06
per_type_v6.csv	flight_list_v6.py	0399ea3	2026-08-10T23:06
pipeline_path_ring_v6.csv	flight_list_v6.py	532610d	2026-08-10T23:57
pipeline_path_v6.csv	flight_list_v6.py	532610d	2026-08-11T00:10
sampler_comparison_v6.csv	sampler_comparison.py	f0f68e4 (dirty)	2026-08-11T08:16
sweep_cone_2025.csv	benchmark_modes.py	d6abb5d	2026-08-10T22:45
sweep_penalty_2025.csv	benchmark_modes.py	d6abb5d	2026-08-10T22:45
sweep_radius_height_2024.csv	benchmark_modes.py	0399ea3	2026-08-10T23:38
sweep_radius_height_2025.csv	benchmark_modes.py	d6abb5d	2026-08-10T22:45
trend_bearing_v6.csv	trend_bearing.py	532610d	2026-08-11T00:17
trend_grid_v6.csv	flight_list_v6.py	0399ea3	2026-08-10T23:32
trend_sweep_2024.csv	trend_sweep.py	e07df0c	2026-08-10T22:38
trend_sweep_2025.csv	trend_sweep.py	9b9404e	2026-08-10T22:29
vertical_measure_v6.csv	vertical_measure.py	0399ea3	2026-08-10T23:42

20 files, 20 traced to a command, 0 unverified.

16.1 Where reproducibility stops

Two boundaries, stated rather than implied.

Ground truth cannot be rebuilt here. `research/reference` is extracted by the `eurocontrol` R package against the PRISME Oracle warehouse, which no run on this cluster can reach. It is an external input, and every figure on this page is measured against it.

The upstream tables are rebuilt only when absent. Reference zones, state vectors and tracks are produced by declared stages, but a stage whose table is already present records its identity rather than rebuilding it — because regenerating the airport zone table would change the candidate set under the whole study. For this version they *were* rebuilt, from the archive upward, because the sampler changed.

17 Limitations

17.1 Parameters that were not swept

Five parameters were held fixed by decision rather than by evidence. Naming them matters: a reader who sees seven tuned values and no caveat will reasonably assume the whole configuration was optimised, and it was not.

Parameter	Where	Held at	Why not swept
Smoothing half-window	`flights.py`	2 samples	Parameterised in config so it is visible and changeable, but a smoothing width interacts with the sample rate rather than the geometry, and the decimation study settled the sample rate separately.
Classification dead band	`flights.py`	none exists	There is no dead band to sweep. The vote margin already plays that role.
Out-of-area border margin	`flights.py`	30 NM	Fixes what counts as "outside the observed area". Changing it redefines the ground truth rather than the method, so a sweep would move the target as well as the aim.
Aerodrome type set	`h3_airport_zones.py`	large + medium	The candidate set. Widening it is a data-volume decision about the zone table, not a detection threshold.

H3 resolution	`config.py`	7 (±0.76 NM)	Changing it invalidates every cached zone table and the `distance_from_center` ring counts that production selects on.

Two of these carry a cost that this study measures rather than waves at, since "not swept" is not the same as "no consequence".

The type set. 980 flights across both roles have a true aerodrome outside the large/medium candidate set (267 at heliports, 713 at small airports). No method on this page can name them, whatever its thresholds, because the aerodrome is not among the candidates. They are counted in every denominator here, so each coverage figure already carries their cost: they are worth about 0.52% of headline coverage.

The border margin. Fixed at 30 NM, it decides which flights are labelled out-of-area. 8.30% of ground-truth departures genuinely begin outside the observed area, and the recommended configuration recognises 7.98% of them — at 89.49% precision, so the label is trustworthy when it appears but is rarely reached. For arrivals the recall is 0.00%: `trend` emits no out-of-area label at all, which is a structural property of the vote rule rather than a tuning choice.

Both costs fall on coverage rather than accuracy, which is the benign direction: with the free out-of-area wins removed, in-area accuracy is 98.58% for departures and 97.99% for arrivals.

17.2 The outstanding check

The endpoint radius and height — the values that actually change what the pipeline ships for departures — have been validated against the pipeline but on **one** period. `trend`, whose tuned values were mostly rejected, is the one measured on two. That is the wrong way round, and it is the first thing a follow-up should fix: build the endpoint candidate cache for a second period and re-run the radius x height grid against it.

Two things limit how much that weakens the recommendation. The departure gain is large — thousands of flights, not tens — so it is not the kind of margin a single sample's noise produces. And the optimum sits on a plateau in both parameters, so even a moderate shift in the argmax would cost little.

17.3 One sample, and what that does and does not license

The parameters are chosen as the argmax over a grid evaluated on a single three-day period. An argmax over hundreds of cells on one sample will always overstate the gain slightly: some of the winning cell's margin is the sample's noise, and it will not reproduce.

That is the reason for the second period, and for reporting the shape of each optimum rather than only its location. A parameter sitting on a broad plateau is a safe recommendation even from one sample; one sitting on a sharp isolated peak is not, however good its score.